



Degassing Unit With Cold Flaring

Programmed By:- Somnath Vishwakarma

Month- September
2026

PLC INPUT AND OUTPUT DIAGRAM

INPUTS

X0	Start_Watering_unit
X1	Stop_Watering_unit
X2	Emg_Switch_Watering_Unit
X3	Reset_Bit_WT
X4	Head_1_Sensor_WT
X5	Head_3_Sensor_WT
X6	Head_5_Sensor_WT
X7	Head_4_Sensor_WT
X20	Head_2_Sensor_WT
X21	Head_6_Sensor_WT
X22	HEAD_05_BYPASS_WT
X23	HEAD_04_BYPASS_WT
X24	HEAD_03_BYPASS_WT
X25	HEAD_02_BYPASS_WT
X26	HEAD_01_BYPASS_WT
X27	HEAD_06_BYPASS_WT
X30	STOPER_BYPASS_DWT
X31	SCM_Motor_Bypass_PB-02
X32	EMG_DWT
X33	DWT_Clamp_relise_Confirmation
X34	GUN_DOWN_CONFIRMATION
X35	START_DWT
X36	HEAD_04_SENSOR_DWT

INPUTS

X37	HEAD_03_SENSOR_DWT
X40	HEAD_06_SENSOR_DWT
X41	HEAD_05_SENSOR_DWT
X42	HEAD_01_SENSOR_DWT
X43	HEAD_02_SENSOR_DWT
X44	STOPER_SENSOR_DWT
X45	RST_C4_DWT
X46	PRESSURE_SWITCH_CLD
X47	EMG_CLD
X50	START_CLD
X51	LAST_SENSOR_WT
X52	LAST_SENSOR_CLD
X53	HEAD_05_SENSOR_CLD
X54	HEAD_01_SENSOR_CLD
X55	HEAD_02_SENSOR_CLD
X56	HEAD_04_SENSOR_CLD
X57	HEAD_03_SENSOR_CLD
X60	STOPER_BYPASS_WT
X61	STOPER_BYPASS_CLD
X62	Spare
X63	WATER_MOTOR_RUNNING
X64	STOPER_SENSOR_WT
X65	Suction_Motor_Bypass_On

OUTPUTS

Y0	Head_2_Clamp_With_Lifter_WT
Y1	Head_3_Gun_WT
Y2	Head_2_Gun_WT
Y3	Head_6_Gun_WT
Y4	Head_6_Clamp_With_Lifter_WT
Y5	Head_4_Clamp_With_Lifter_WT
Y20	Head_1_Gun_CLD
Y21	Head_1_Clamp_With_Lifter_WT
Y22	Head_4_Gun_WT
Y23	Head_3_Clamp_With_Lifter_WT
Y24	Stoper_Pre_Watering_unit_WT
Y25	Head_1_Gun_WT
Y26	Head_5_Clamp_With_Lifter_WT
Y27	Head_5_Gun_WT
Y30	Head_3_Clamp_With_Lifter_DWT
Y31	Head_2_Clamp_With_Lifter_DWT
Y32	Head_6_Clamp_With_Lifter_DWT
Y33	Stoper_Pre_Dewatering_unit_DWT
Y34	Head_5_Clamp_With_Lifter_DWT
Y35	Head_1_Clamp_With_Lifter_DWT
Y36	Head_4_Clamp_With_Lifter_DWT
Y37	Green_Light_DWT

OUTPUTS

Y40	Head_4_Gun_CLD
Y41	Head_1_Clamp_With_Lifter_CLD
Y42	Stoper_Pre_Watering_unit_CLD
Y43	Head_3_Gun_CLD
Y44	Head_5_Clamp_With_Lifter_CLD
Y45	Head_4_Clamp_With_Lifter_CLD
Y46	Head_2_Clamp_With_Lifter_CLD
Y47	Head_5_Gun_CLD
Y50	Head_2_Gun_CLD
Y51	Head_3_Clamp_With_Lifter_CLD
Y52	Green_Light_CLD
Y53	RED_LIGHT_CLD
Y54	RED_LIGHT_DWT
Y55	AIR_ROV_DWT
Y56	Filling_Motor_WT_RELAY
Y57	SUCTION_MOTOR_DWT_RELAY
Y60	Gun_Air_ON/OFF_DWT
Y61	END_Stoper_DWT
Y62	ROV_CLD
Y63	Green_Light_WT
Y64	Yellow_Light_for_byypass_condition
Y65	Red_Light_WT

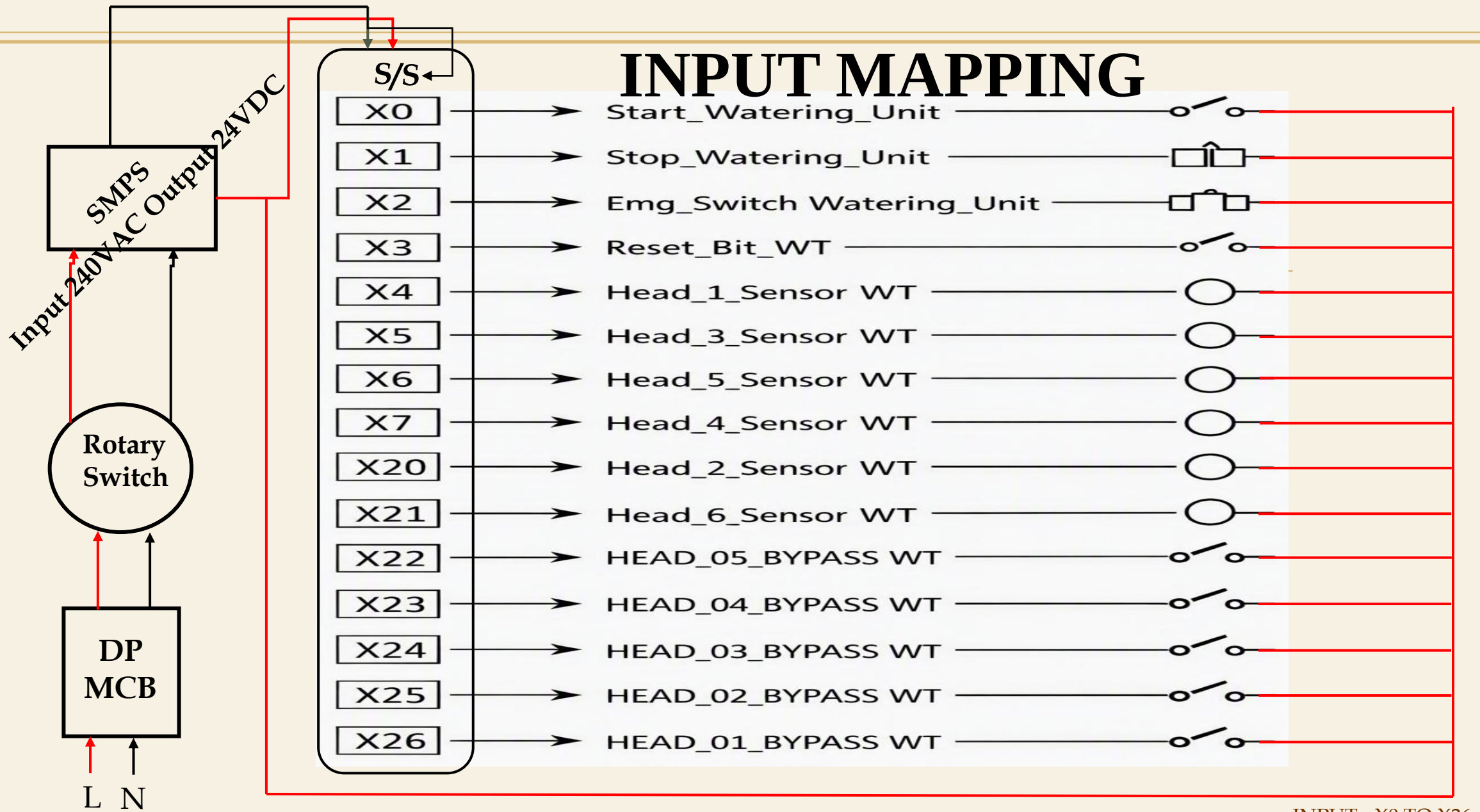


SLD (*Single Line Diagram*)

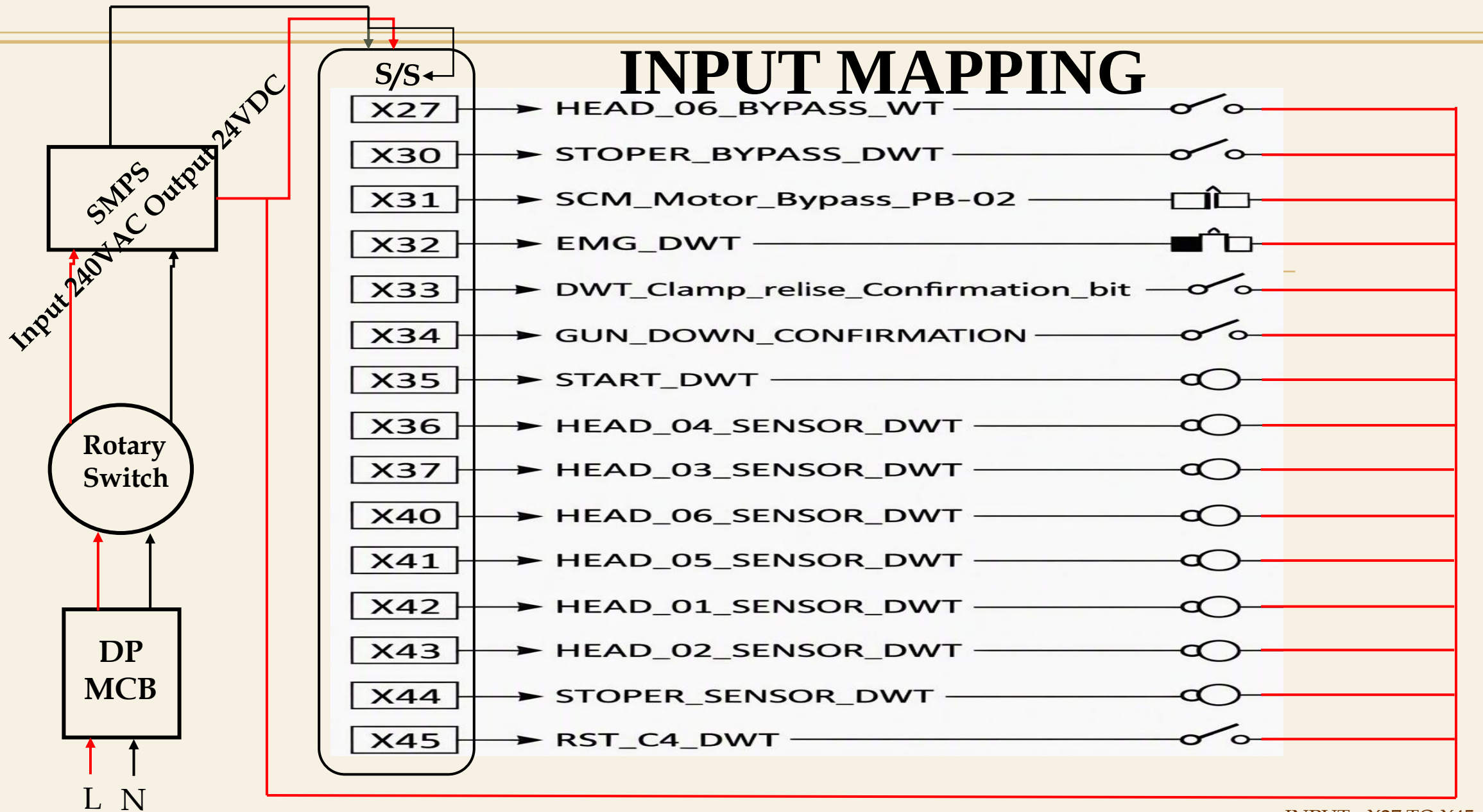
SLD Done By:- Somnath Vishwakarma

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INPUT MAPPING

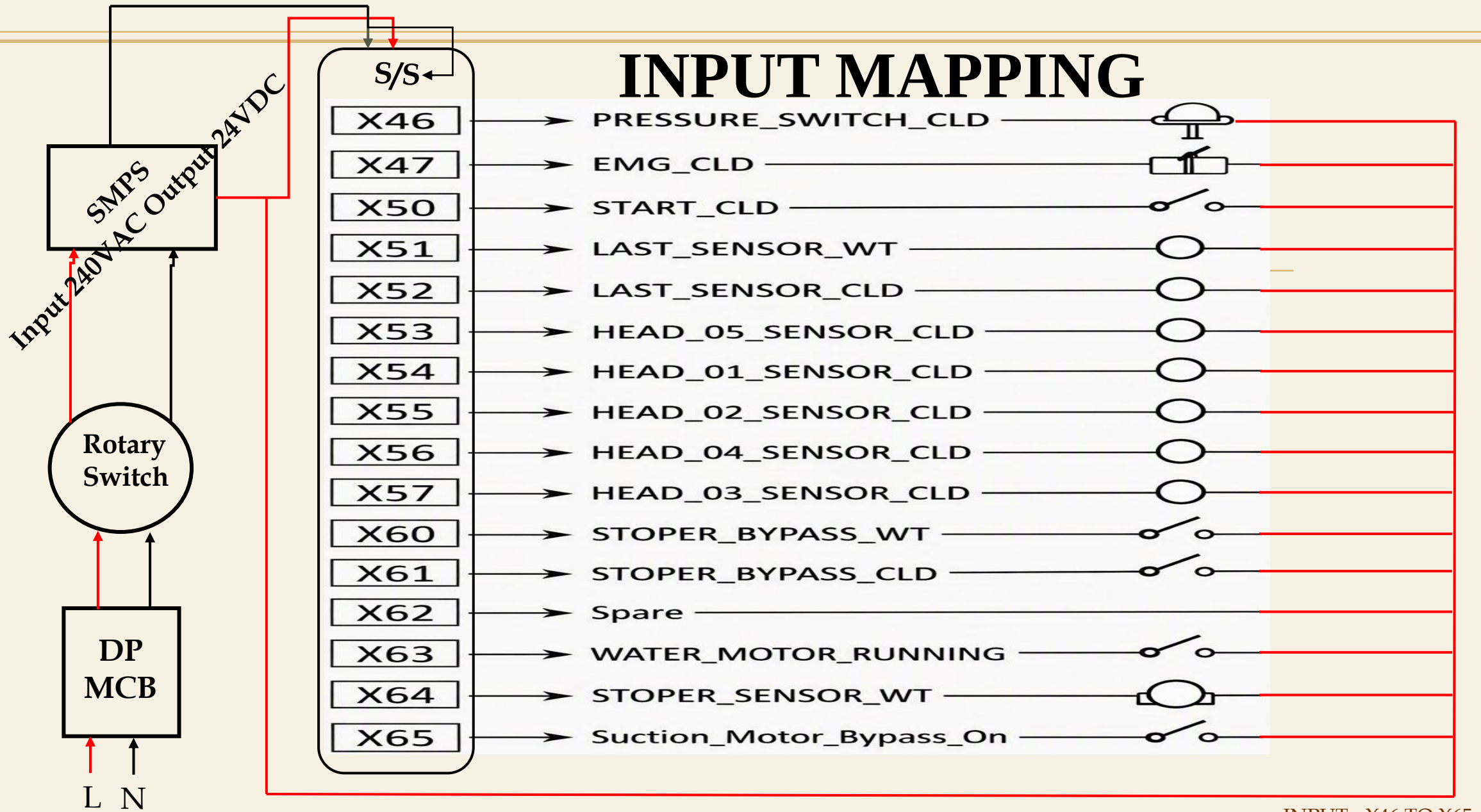


INPUT MAPPING



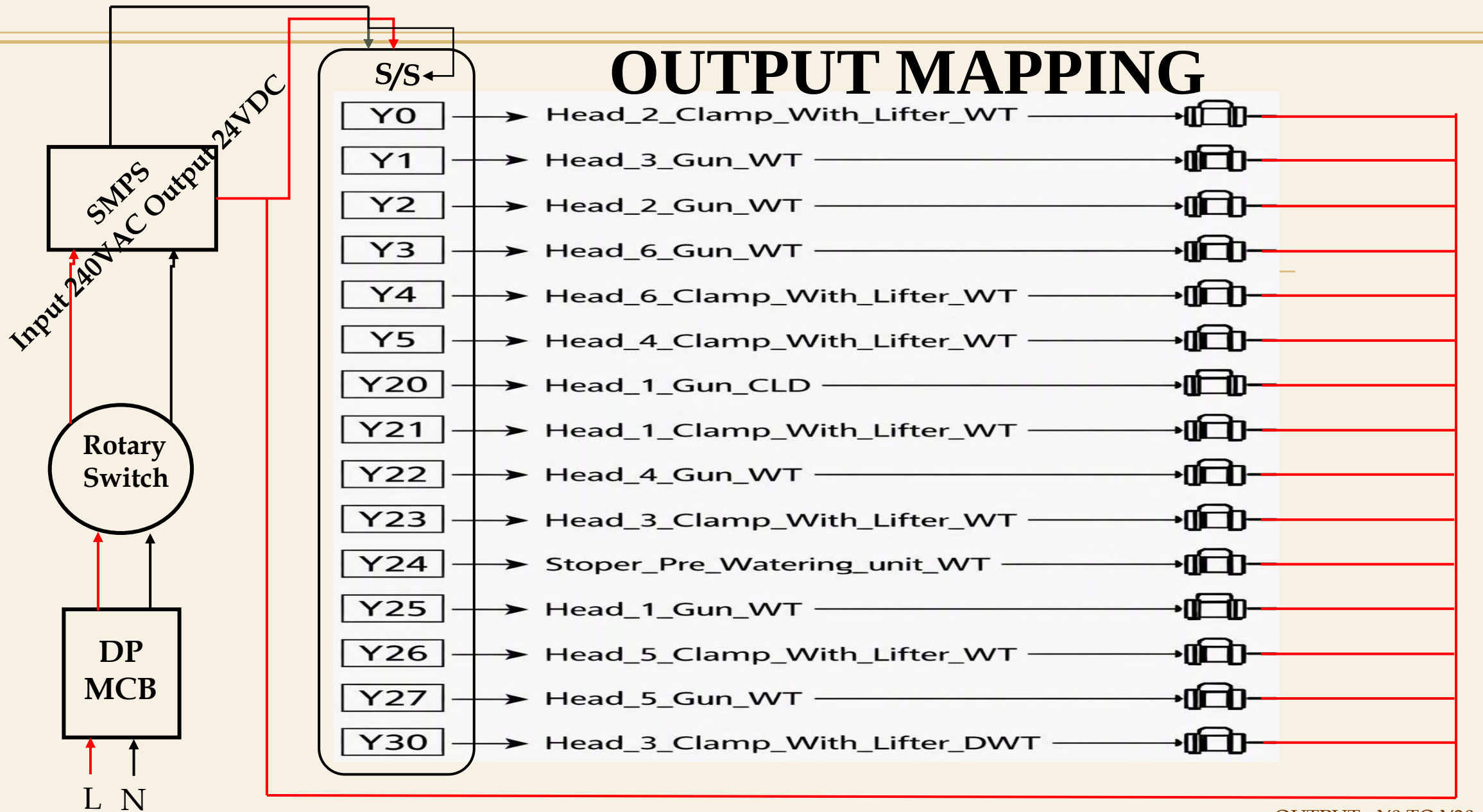
INPUT:- X27 TO X45

INPUT MAPPING



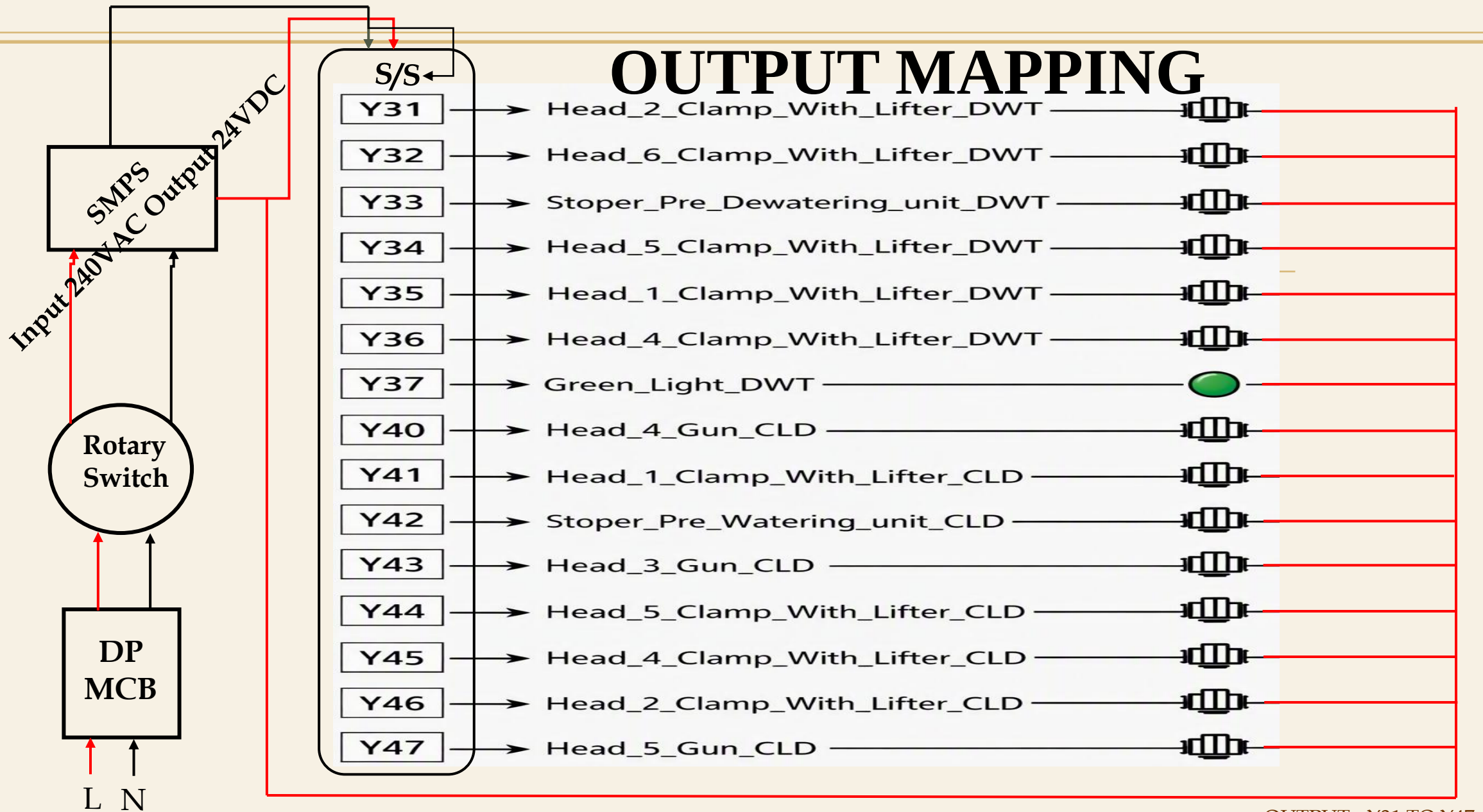
INPUT:- X46 TO X65

OUTPUT MAPPING



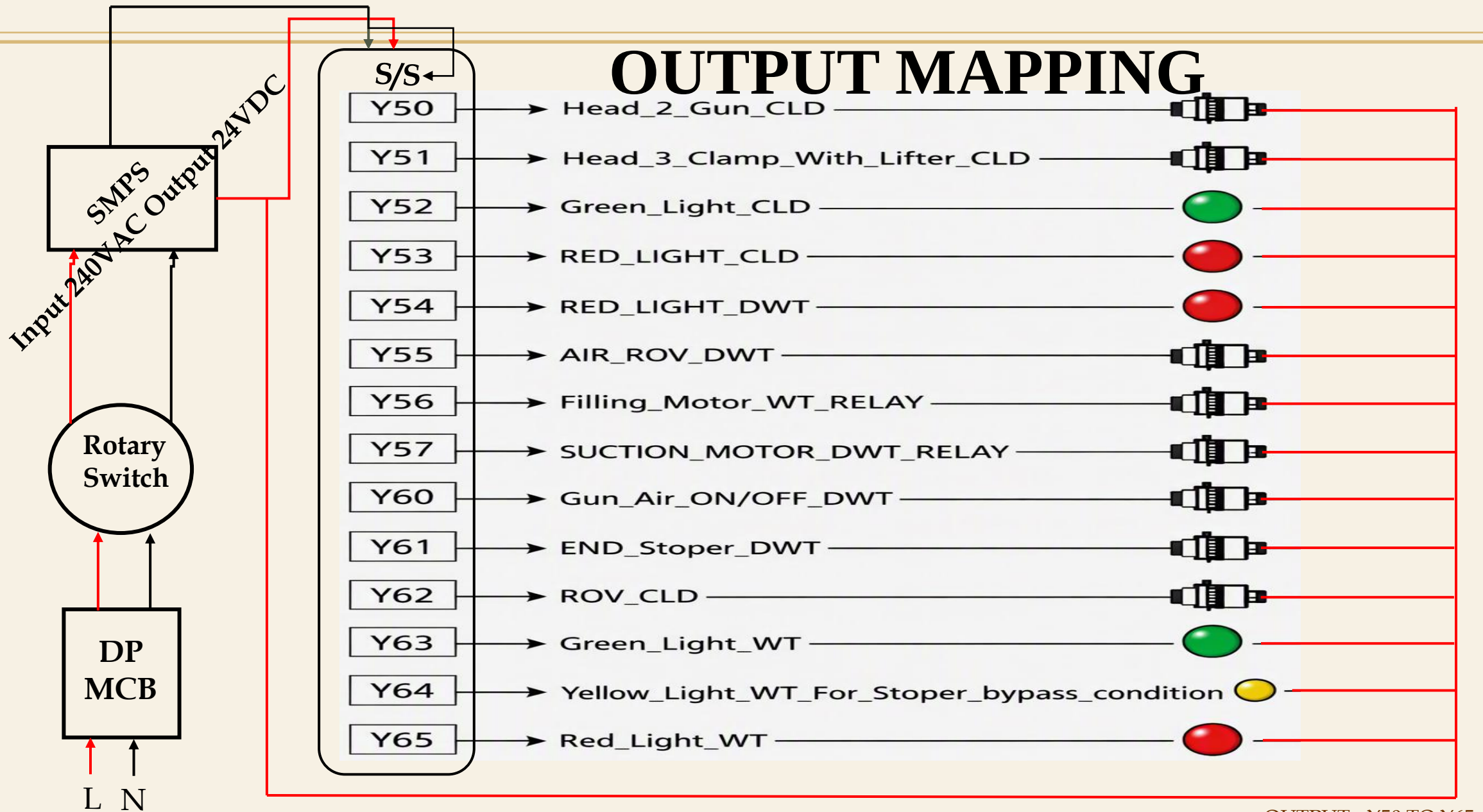
OUTPUT:- Y0 TO Y30

OUTPUT MAPPING



OUTPUT:- Y31 TO Y47

OUTPUT MAPPING



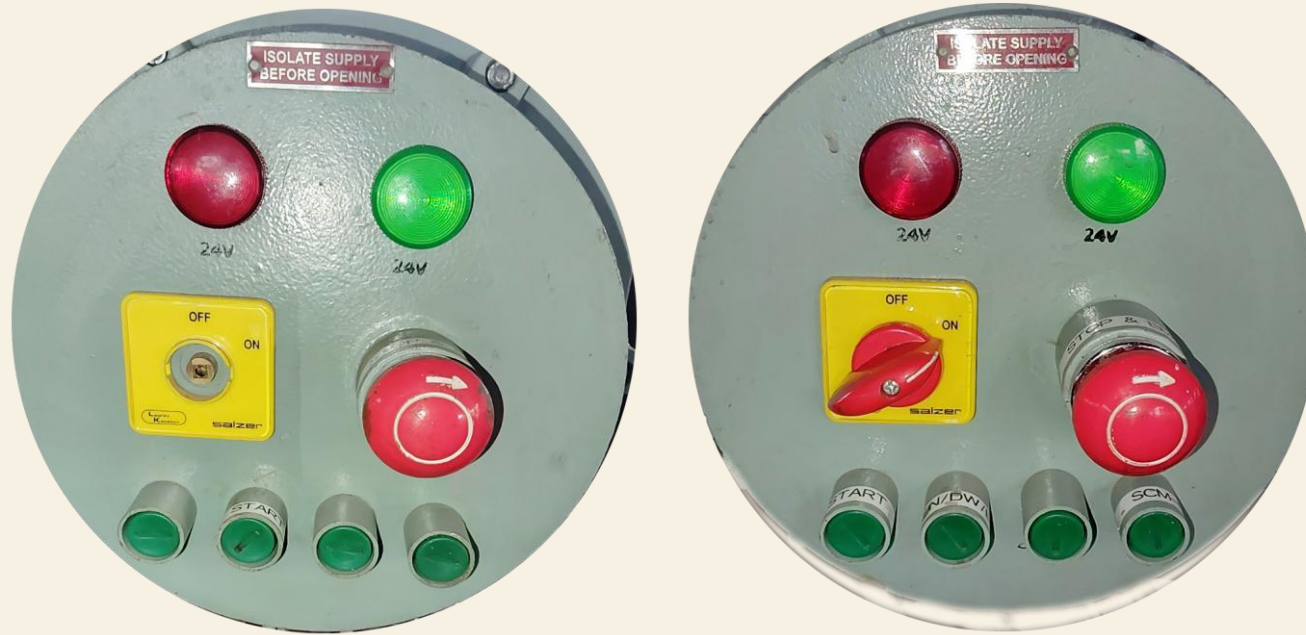
OUTPUT:- Y50 TO Y65



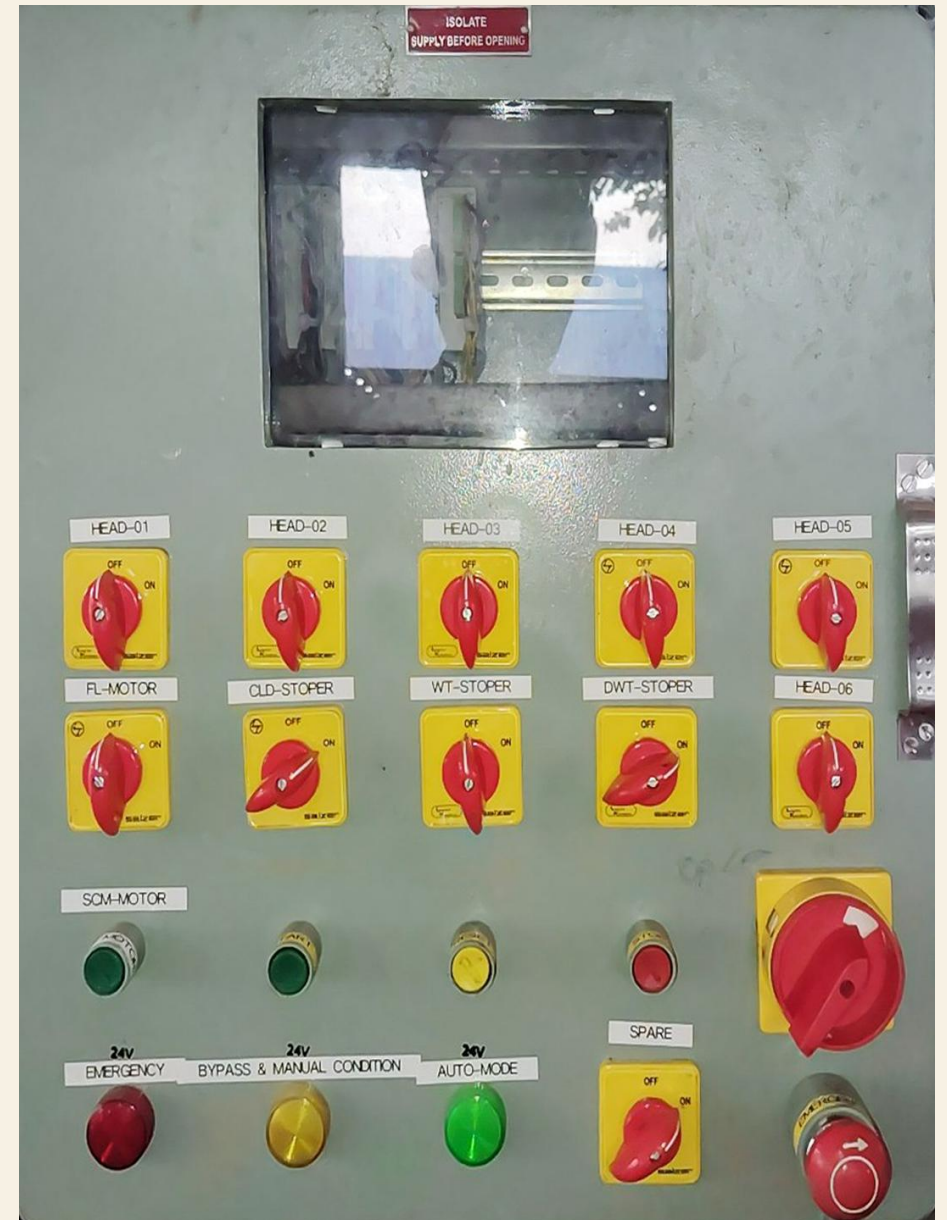
Panel Details (*Make :-* SAIEX)

Panel Done By:- Somnath Vishwakarma

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MAKE:- SAIEX



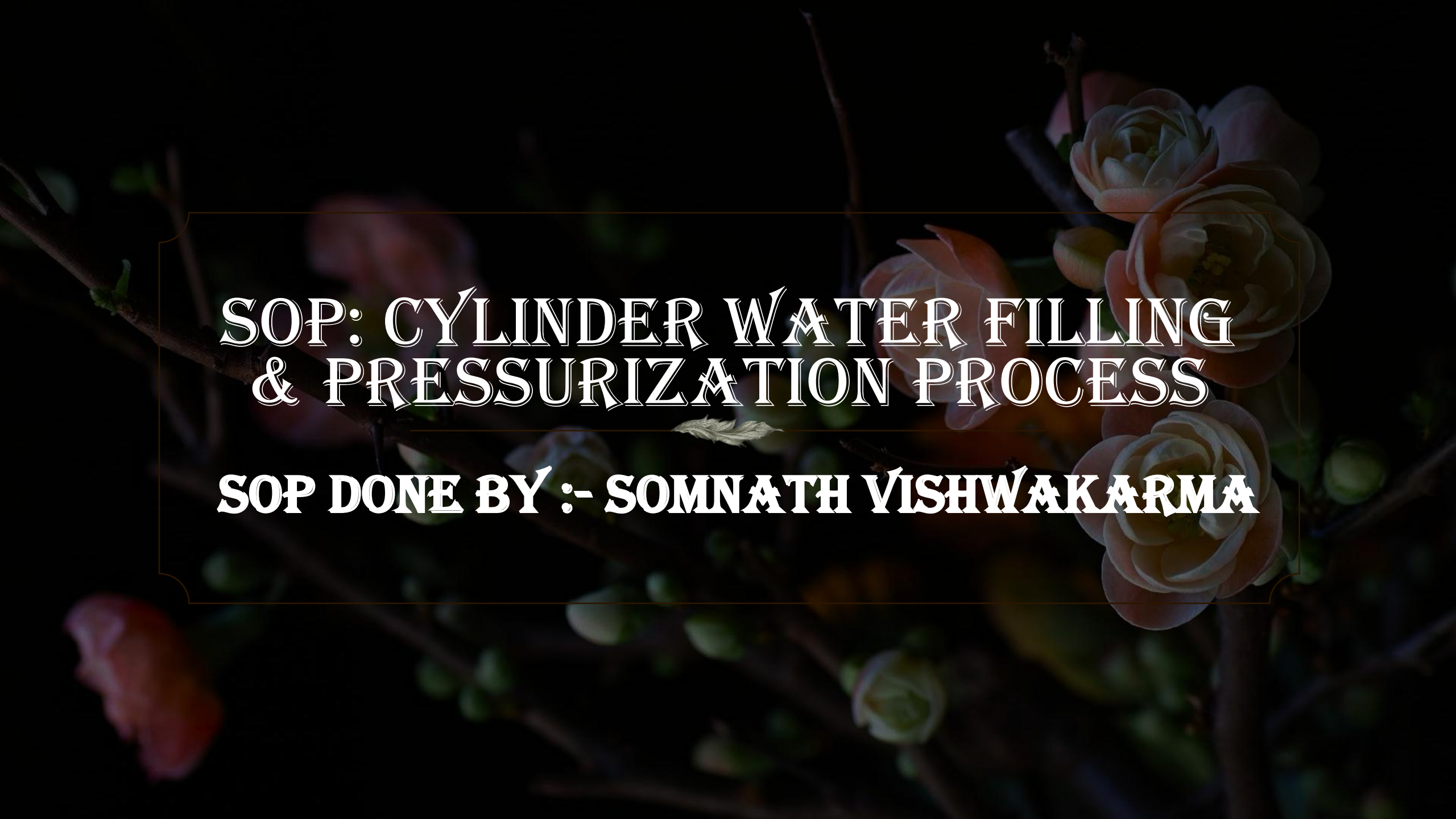


CIMFR DOCUMENT

PESO_1 DOCUMENT

PESO_2 DOCUMENT

Main Panel:- Degassing Shed



SOP: CYLINDER WATER FILLING & PRESSURIZATION PROCESS

SOP DONE BY :- SOMNATH VISHWAKARMA

Cold Flaring Machine Start-Up Procedure

01. Check Stoper Bypass Condition

- Before starting the cold flaring machine, verify the stopper bypass condition.
- If the stopper bypass is **ON**, the machine will not start and the green indicator light will remain off.
- Remove the bypass condition, then press **Start**.
- Once cleared, the green LED on the front panel will glow continuously, indicating the machine is ready for automatic operation.

02. Preparation Before Start

- Stop the conveyor.
- Remove all cylinders from inside the machine.
- First start the machine, then start the chain conveyor.

Cold Flaring Machine Start-Up Procedure

•03. Automatic Operation Sequence

- Cylinders will enter the sensor range one by one.
- The system will automatically clamp the cylinder, lower the gun, and begin flaring.
- If vapor is present inside the cylinder, it will be automatically released.
- Once vapor is cleared, cylinders will be processed one by one in sequence.

•04. Important Notes

- Do not stop the chain conveyor while the machine is running in auto mode.
- If liquid LPG enters the machine, removal may take longer and may not be possible properly.
- In such cases, press the Emergency (EMG) Switch, remove all cylinders, and restart the machine following the above procedure.

Water Filling Unit. Start-Up Procedure

- **01. Check Stoper Bypass Condition**
- Setup
- Ensure the machine is not in bypass mode before starting.
- Verify stoper bypass condition
- If bypass is ON, machine will not start and green light will remain off
- Remove bypass condition to proceed
- **02. Prepare Machine and Conveyor**
- Get the system ready before pressing start.
- Stop the conveyor
- Remove all cylinders from inside the machine
- Confirm machine is clear before operation
- **03. Start Machine and Conveyor**
- Initiate the automatic sequence.
- Press the **Start** button
- Turn on the chain conveyor
- Cylinders will enter sensor range one by one
- **04. Automatic Clamping and Lifting**
- System clamps and lifts cylinders automatically.
- Cylinder detected by sensor
- Automatic clamp engages
- Lifter raises cylinder
- All heads move downward

Water Filling Unit. Start-Up Procedure

- **05. Manual Alignment if Needed**

- Safety
- Correct cylinder alignment before filling.
- If any head is not properly aligned, use manual alignment option
- Manual buttons are provided on the front side of the main panel

- **06. Start Filling Motor**

- Begin the filling process once alignment is confirmed.
- Press **Start Filling Motor**
- Filling begins automatically
- Gun moves up and lifter lowers cylinders one by one

Dewatering Unit. Start-Up Procedure

• **01. Check Stoper Bypass Condition**

- Setup
- Ensure the machine is not in bypass mode before starting.
- Verify stoper bypass condition.
- If bypass is ON, machine will not start and green light will remain off.
- Remove bypass condition to proceed.
- **02. Prepare Machine**
- Stop conveyor and clear cylinders before initiating operation.
- Stop the conveyor.
- Remove all cylinders from inside the machine.
- Insert pipe into the cylinder.

• **03. Start Machine and Conveyor**

- Begin automatic sequence with proper cylinder detection.
- Press the **Start** button.
- Turn on the chain conveyor.
- Cylinders enter sensor range one by one.
- Automatic clamp and lifter engage with each cylinder.
- **04. Tighten Gun and Pipe**
- Safety
- Ensure gun and pipe are properly tightened for effective de-watering.
- Gun and pipe tighten inside cylinder.
- If not properly tight, water blanking will fail.
- Use manual hand lever to adjust and tighten all heads.

Dewatering Unit. Start-Up Procedure

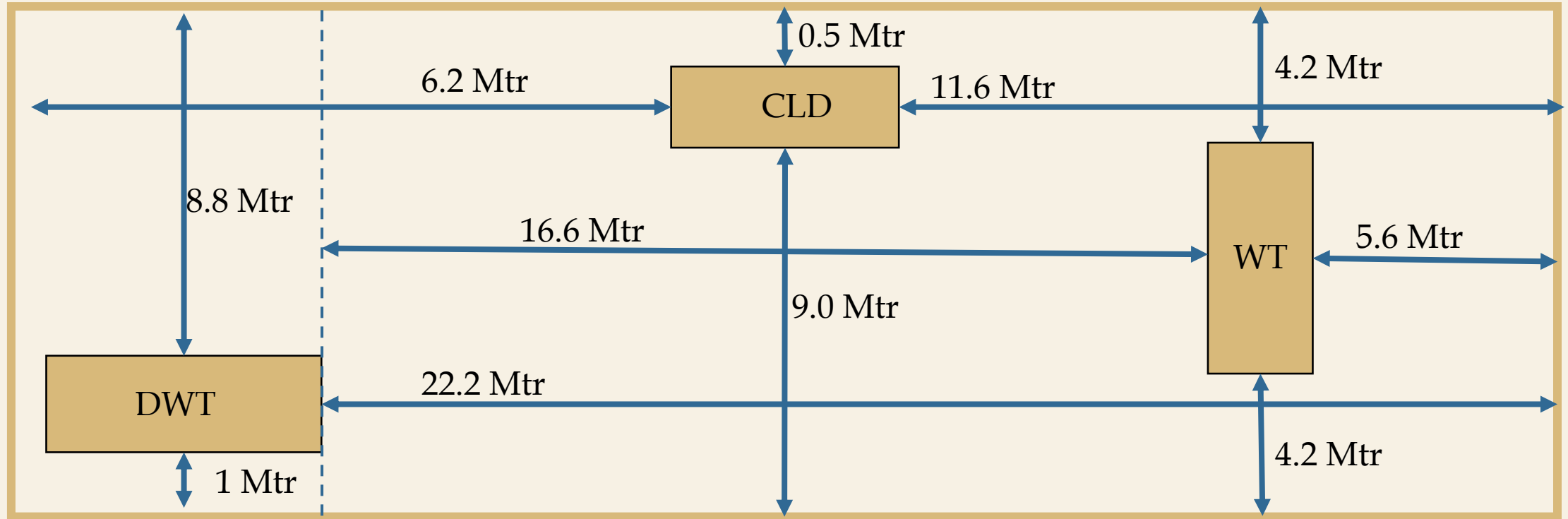
- **05. Confirm Gun Operation**

- Start gun confirmation and allow automatic blanking.
- Press **Start Gun Confirmation**.
- Cylinder blanking motor runs automatically.
- Use hand lever to lift gun up after blanking.
- Remove gun and pipe.

- **06. Release Cylinders**

- Detach equipment and release cylinders sequentially.
- All gun and pipe are de-latched.
- Press **Release Button**.
- Cylinders are automatically released one by one.

LAYOUT



Measurement as Lumpsum Data.



Thank You!

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